

CO5 – AT3 DAA MOCK INTERVIEW REPORT

Student Name: Murali Krishnan N

Course: B.Tech Artificial Intelligence and Data Science

Institution: SIMATS Engineering

Interview Summary

The CO5 – AT3 Mock Interview was conducted on Design and Analysis of Algorithms (DAA). The interview consisted of 10 questions covering Backtracking, State-Space Trees, Pruning, N-Queens, Subset Sum, Hamiltonian Cycle, Graph Coloring, P/NP/NP-Hard/NP-Complete, Greedy Algorithms, and Exact vs Heuristic Algorithms.

Question-Wise Performance

1. Backtracking – Good

Explained using the example of solving a maze by trying different paths.

2. State-Space Tree – Excellent

Correctly explained root, branches, nodes, and solution nodes.

3. Pruning – Excellent

Correctly explained stopping paths that cannot lead to a valid solution.

4. N-Queens – Excellent

Correctly explained row, column, and diagonal restrictions.

5. Subset Sum – Excellent

Correctly explained finding a subset equal to a target sum.

6. Hamiltonian Cycle – Excellent

Correctly explained visiting every vertex once and returning to the start.

7. Graph Coloring – Excellent

Correctly explained that adjacent vertices cannot have the same color.

8. P, NP, NP-Hard, NP-Complete – Excellent

Clearly differentiated all four complexity classes.

9. Greedy Algorithm – Good

Correct definition; an example was initially missing.

10. Exact vs Heuristic – Excellent

Clearly explained optimal versus near-optimal solutions.

Evaluation

Criterion	Performance	Marks
Technical Knowledge	Very Good	23/25
Problem Solving	Very Good	23/25
Analytical Thinking	Good	13/15
Communication	Very Good	14/15
Confidence	Good	8/10

Response to Questions	Excellent	9/10
TOTAL	VERY GOOD	90/100

Technical Strengths

- Good understanding of Backtracking concepts.
- Strong knowledge of N-Queens, Subset Sum, and Graph problems.
- Clear understanding of P, NP, NP-Hard, and NP-Complete.
- Good knowledge of Exact and Heuristic Algorithms.
- Answers were mostly accurate and concise.

Areas for Improvement

- Give an example whenever the interviewer asks for one.
- Speak in complete sentences during the interview.
- Add a little more explanation to demonstrate deeper understanding.
- Improve confidence while answering longer questions.

Overall Performance

Overall Level: Very Good

Total Marks: 90/100

Murali Krishnan N demonstrated strong fundamental knowledge of Design and Analysis of Algorithms. Most answers were correct, clear, and concise. With slightly more detailed explanations and examples, the student can achieve an excellent performance in the actual interview.

Final Remark

Murali Krishnan N performed very well in the CO5 – AT3 DAA Mock Interview. Keep practicing with confidence and include simple examples in your answers whenever possible.